Principle of Drug Actions

Drug Dose Administration

Pharmaceutical

Disintegration of Drug

Pharmacokinetics

Absorption/distribution metabolism/excretion

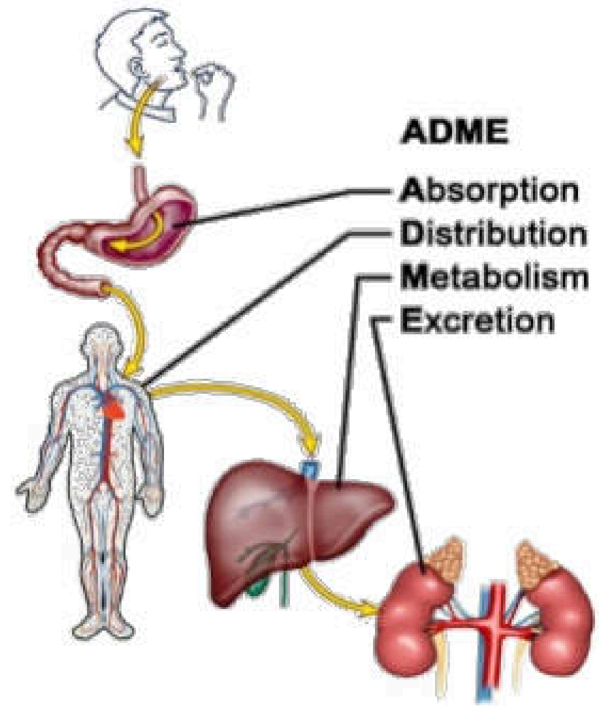
Pharmacodynamics

Drug/Receptor Interaction

Pharmacotherapeutics

Drug Effect or Response

Pharmacokinetics



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Principles of Drug action

- **Stimulation:** Enhancement of the level of a specific biological activity (usually already ongoing physiological process). E.g. adrenaline stimulates heart rate.
- **Depression:** Diminution of the level of a specific biological activity (usually already ongoing physiological process). E.g. barbiturate depress the CNS.
- **Replacement:** Replacement of the natural hormones or enzymes (any substance) which are deficient in our body. E.g. insulin for treating diabetes.
- **Cytotoxic** action: Toxic effects on invading micro organisms or cancer cells.

Drugs act at four different levels:

- **Molecular**: protein molecules are the immediate targets for most drugs. Action here translates into actions at the next level.
- **Cellular**: biochemical and other components of cells participate in the process of transduction.
- **Tissue**: the function of heart, skin, lungs, etc., is then altered.
- **System**: the function of the cardiovascular, nervous, gastrointestinal system, etc., is then altered.

The mechanism of drug action at the four different levels is also illustrated by rifampicin, even though this drug acts on bacterial rather than human tissue. Rifampin is an effective drug in the treatment of tuberculosis:

- At the molecular level, rifampin binds to (and blocks the activity of) ribonucleic acid (RNA) polymerase in the mycobacterium responsible for tuberculosis.
- At the cellular level, rifampin inhibits mycobacterial RNA synthesis and thereby kills the mycobacterium.
- At the tissue level, rifampin prevents damage to lung tissue arising as a result of mycobacterial infection.
- At the system level, rifampin prevents the loss of lung function caused by the mycobacterial infection.

Pharmacodynamics:

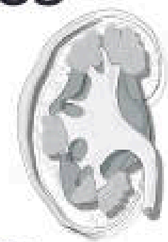
1. How the drugs act on the body?
2. The mechanism of action of drug and its effects.
3. The mechanism of action represents the interaction between drug molecules and biological structures of the organism.
4. **The effect** represents the final results from the drug action, can be observed and measured, but not the action.
5. Site of drug action:
   - They can be divided into: **specific and non-specific**

Targets for drug actions

Non-specific

- Osmotic diuretics

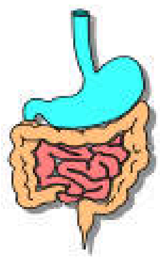
  Mannitol



- Osmotic laxative

  Duphalac

  $MgSO_4$



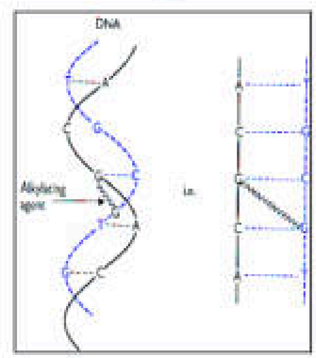
- Antacids (antacids)

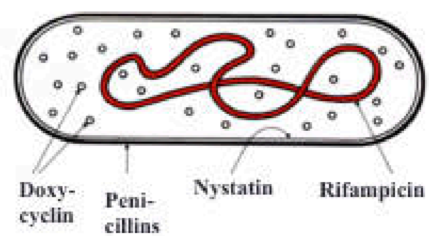
  $NaHCO_3$

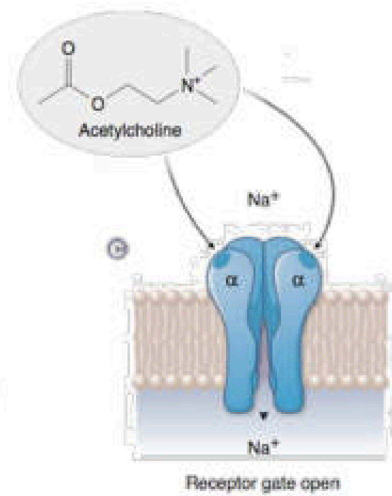
Specific

- DNA
- Microbial organelles
- Target macroproteins

  ion channel, receptor, transporter, enzyme, etc.







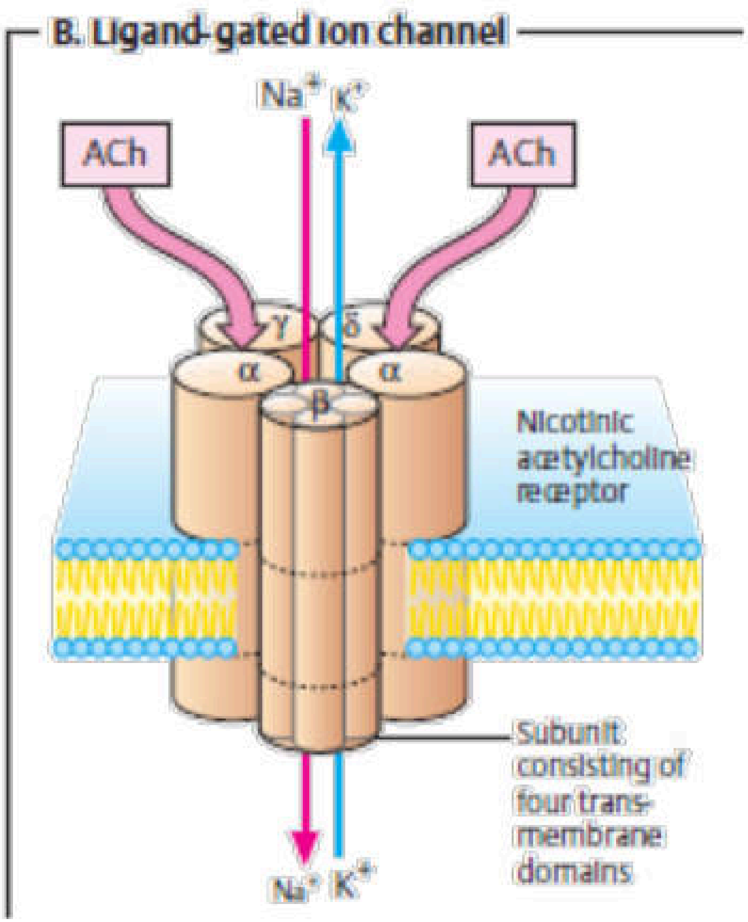
Pharmacodynamic concepts

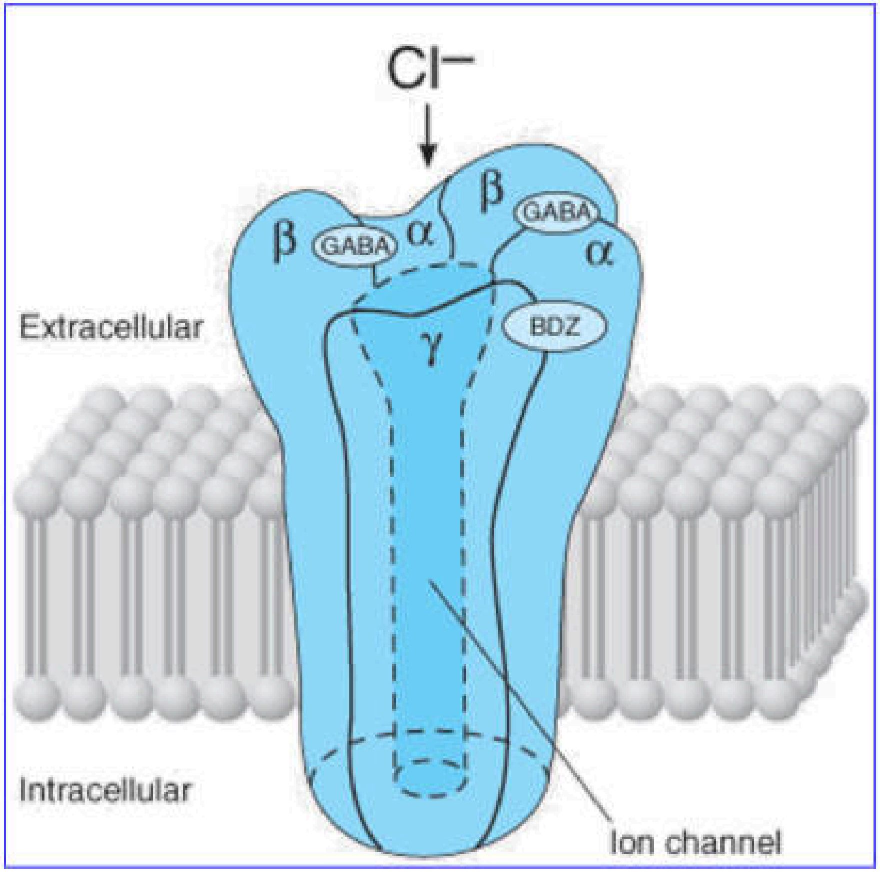
Receptors:

- Specific molecules in a biologic system with which drugs interact to produce changes in the function of the system.
- Receptors must be selective in their ligand-binding characteristics, recognition site for a drug is the specific binding region of the macromolecule and has a high and selective affinity for the drug molecule.
- Receptors also must be modified as a result of binding an agonist molecule.
- The majority of the receptors characterized to date are proteins; a few are other macromolecules such as DNA.
- The interaction of a drug with its receptor is the fundamental event that initiates the action of the drug.

1. ionotropic receptors

Receptors that regulate membrane ion channels may directly cause the opening of an ion channel (eg, ACh at the nicotinic receptor) or modify the ion channel's response to other agents (eg, benzodiazepines at the GABA channel).





- The result is a change in transmembrane electrical potential.

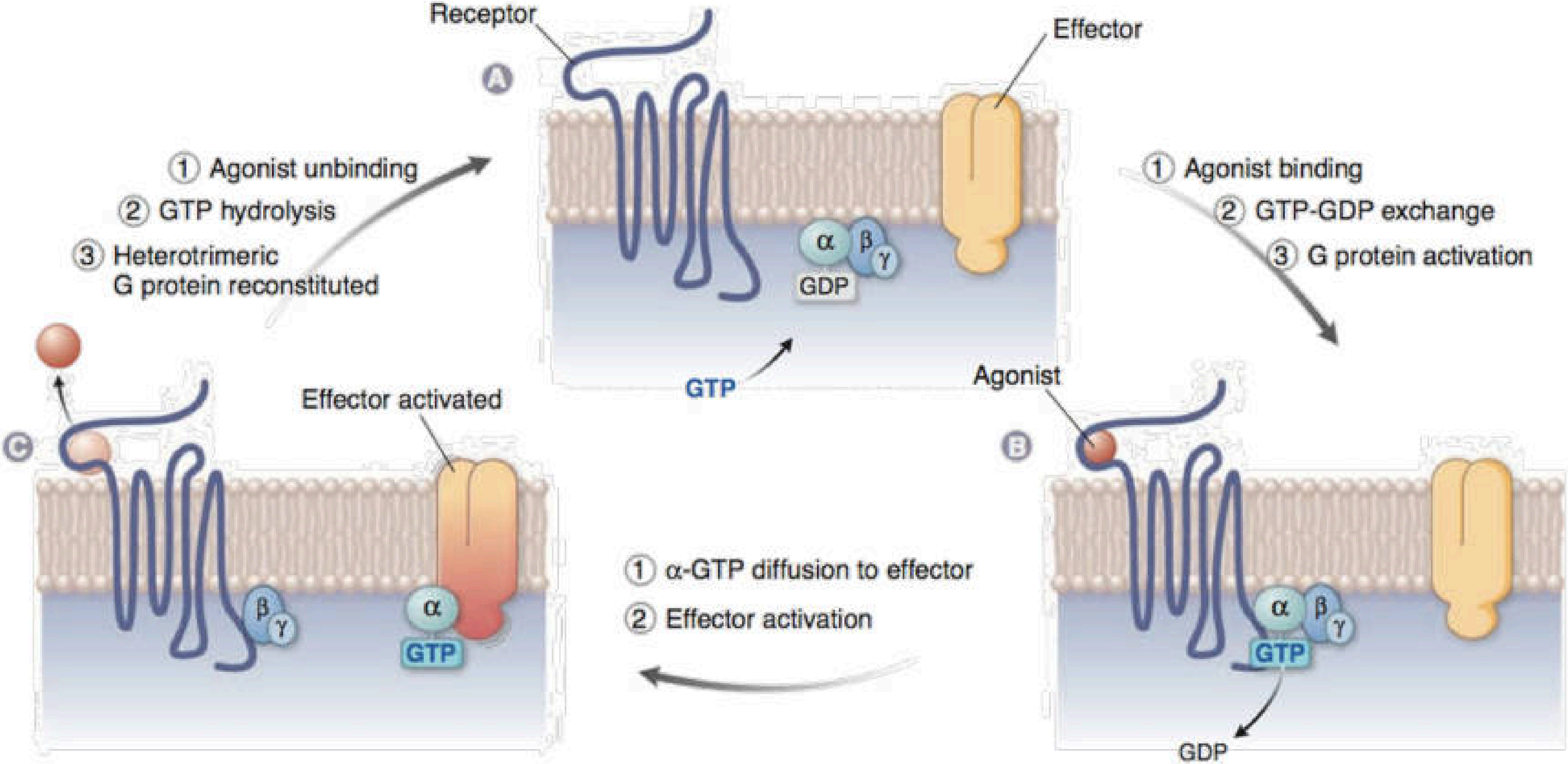
2. Metabotropic receptors

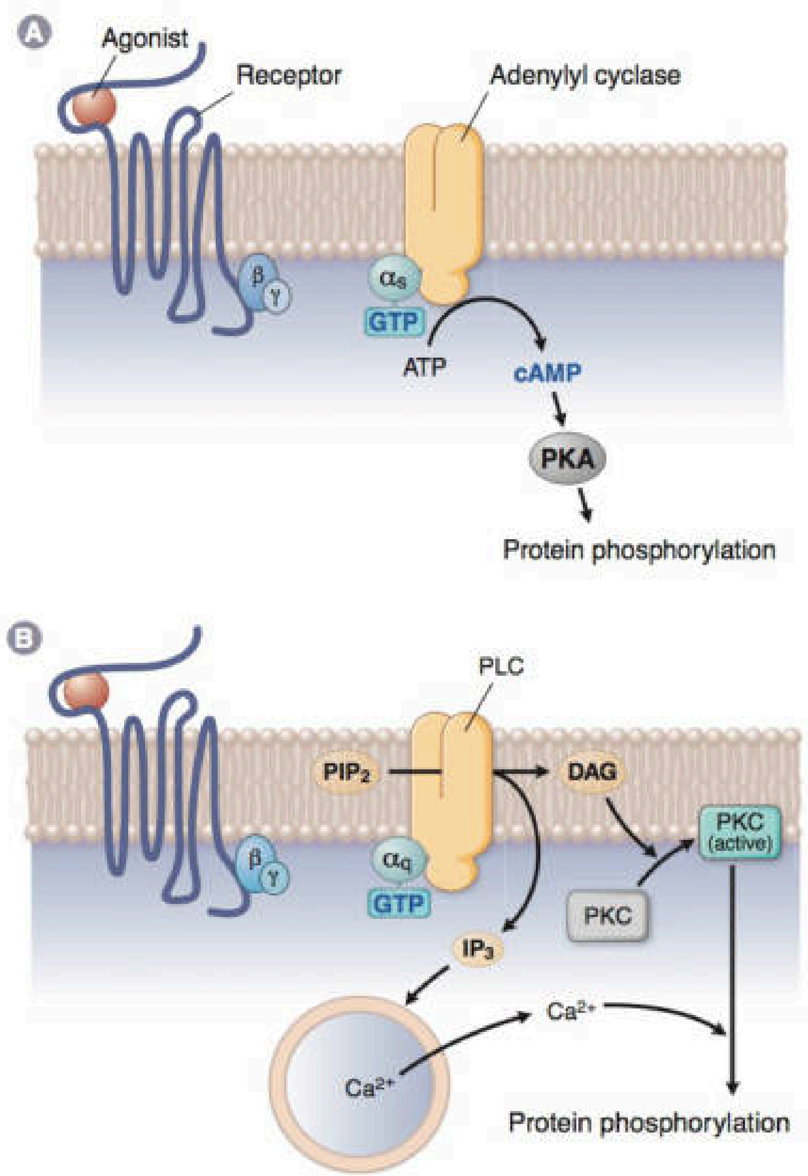
Receptors linked to effectors via G proteins:

- A very large number of drugs bind to receptors that are linked by coupling proteins to intracellular or membrane effectors.
- The best defined examples of this group are the sympathomimetic drugs, which activate or inhibit adenylyl cyclase by a multistep process:
  - activation of the receptor by the drug results in activation of G proteins that either stimulate or inhibit the cyclase.

Receptor-mediated activation of a G protein and the effector interaction

- Single polypeptide threaded back and forth resulting in 7 transmembrane a helices
- G protein attached to the cytoplasmic side of the membrane (functions as a switch)





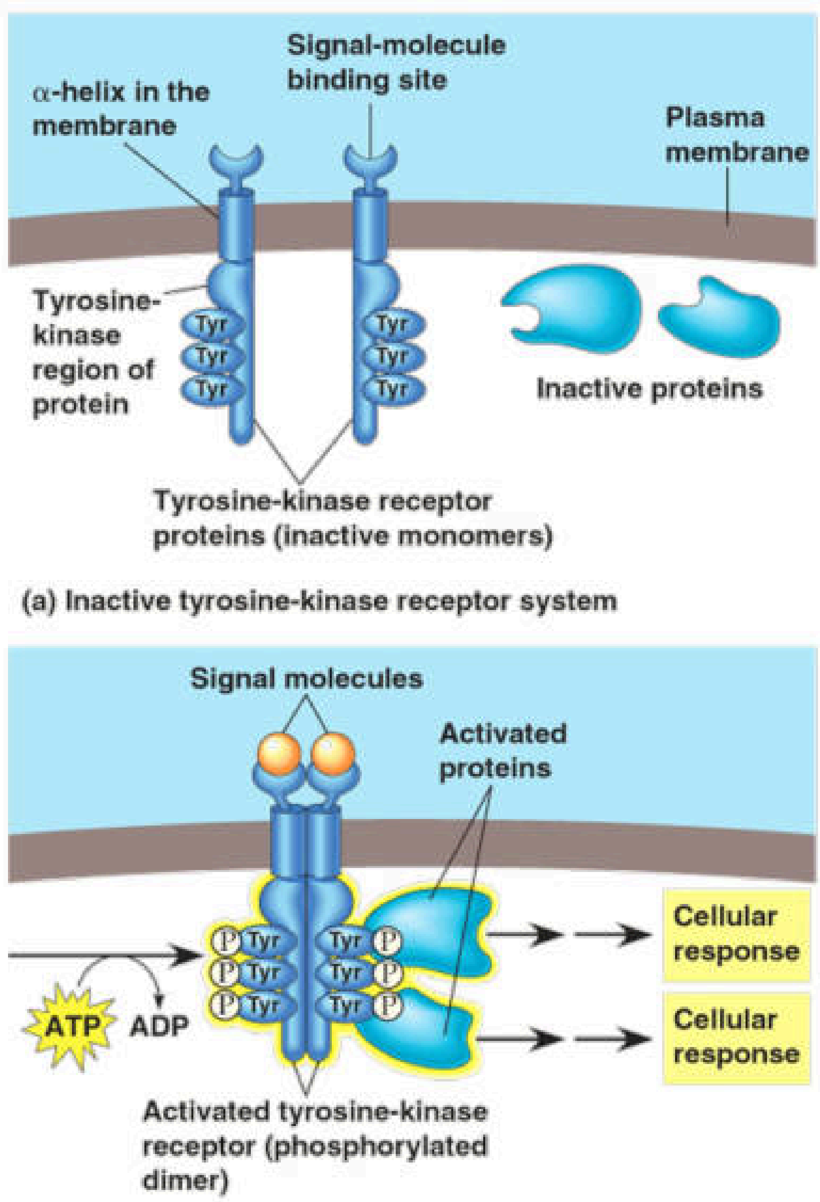
> 20 types of G proteins have been identified; three of the most important are listed in this table.

G PROTEIN	ACTIONS
G-stimulatory (G_s)	Activates Ca^{2+} channels, activates adenylyl cyclase
G-inhibitory (G_i)	Activates K^+ channels, inhibits adenylyl cyclase
G_o	Inhibits Ca^{2+} channels
G_q	Activates phospholipase C
$G_{12/13}$	Diverse ion transporter interactions

3. Receptors located on membrane-spanning enzymes (enzyme-linked)

- Drugs that affect membrane-spanning enzymes combine with a receptor on the extracellular portion of enzymes and modify their intracellular activity.
- For example, insulin acts on a tyrosine kinase that is located in the membrane.
- The insulin receptor site faces the extracellular environment and the enzyme catalytic site is on the cytoplasmic side.
- When activated, the receptors dimerize and phosphorylate specific protein substrates.

Receptor tyrosine kinases: RTK



(a) Inactive tyrosine-kinase receptor system

(b) Activated system

Ligands bind to both receptors

The two receptor polypeptides aggregate forming a dimer

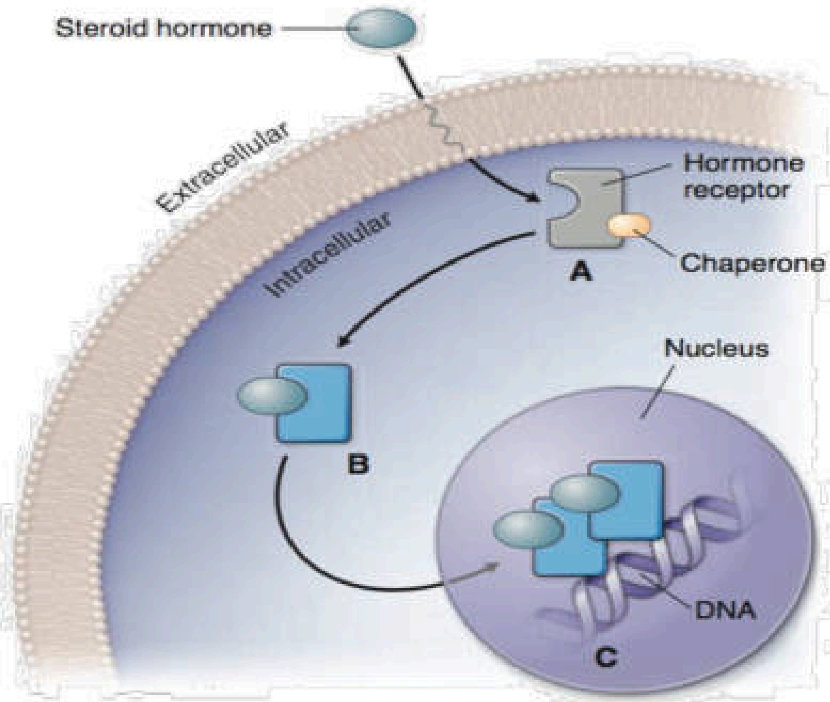
Activates the tyrosine-kinase parts of the dimer

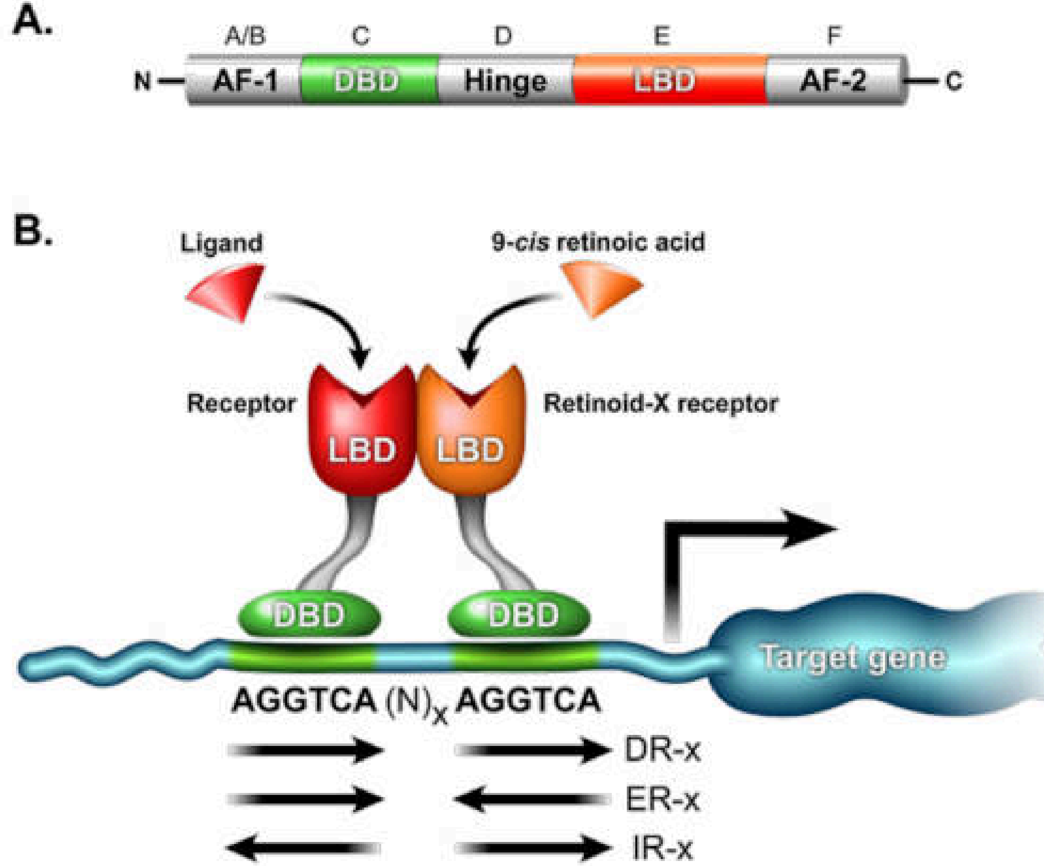
Each phosphorylates (using ATP) the tyrosines on the tail of the other polypeptide

Receptor proteins are now recognized by relay proteins inside the cell

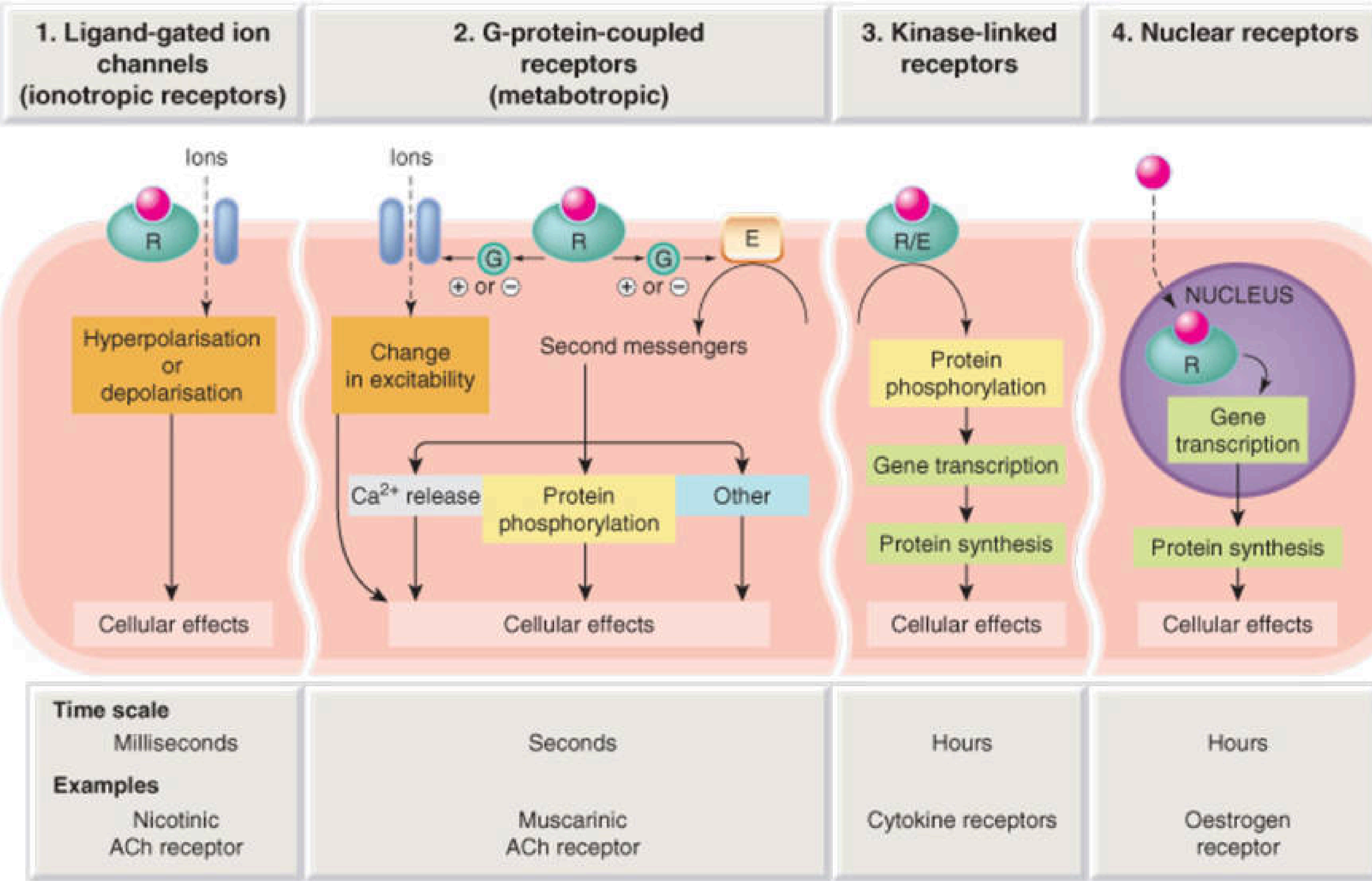
Relay proteins bind to the phosphorylated tyrosines (may activate 10 or more different transduction pathways)

4. Receptors that are intracellular:





- Some drugs, especially more lipid-soluble or diffusible agents (eg, steroid hormones, nitric oxide) may cross the membrane and combine with an intracellular receptor that affects an intracellular effector molecule.

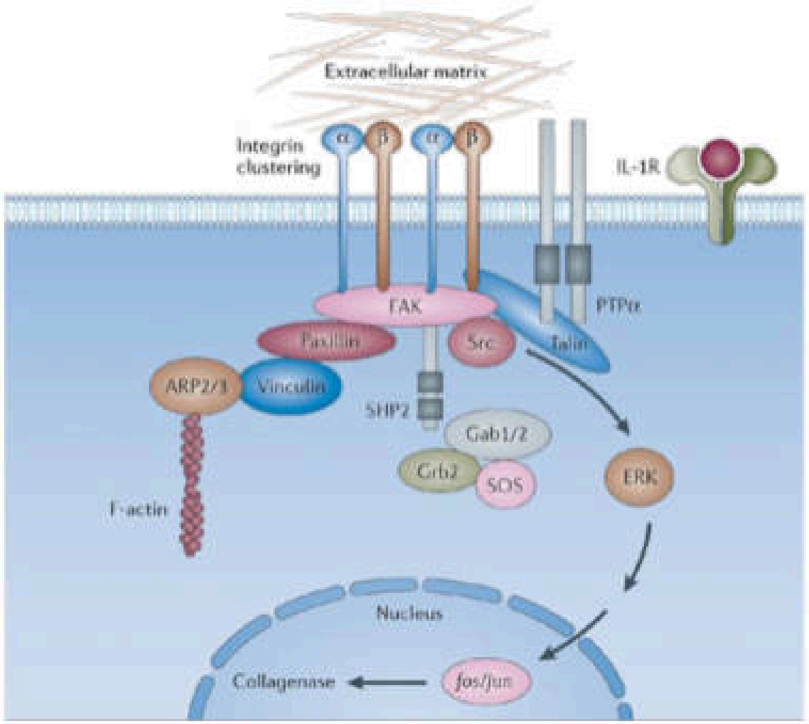


Types of receptor-effector linkage (R = receptor; G = G-protein; E = enzyme)

Other Macromolecules: Drugs Can Bind

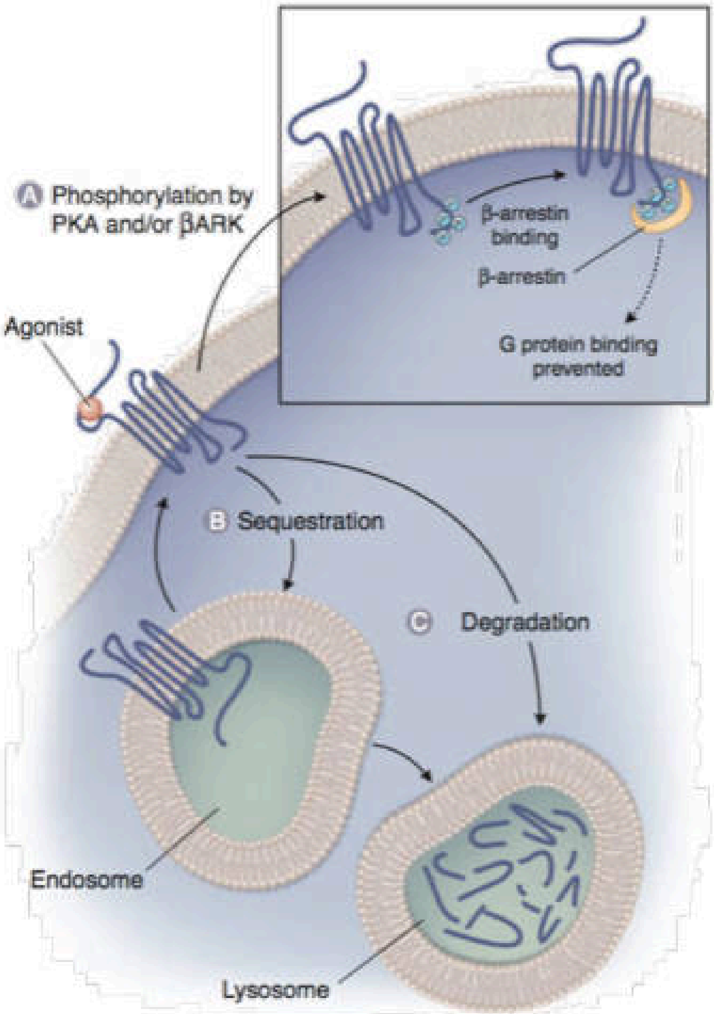
- Enzymes
- Ribosomes
- Tubulin
- Adhesion
- DNA or RNA
- Transporters
- Others

Drug	Action on enzyme
Galantamine	(−) ACh-esterase
Digoxin	(−) Na^+/K^+-ATP-ase
Aspirin	(−) COX-1/COX-2



Drug Specificity: Binding to Multiple Receptor

- Many drugs bind to more than one receptor type and have effects due to activity at all those receptors.
  - If a drug has activity at > one receptor, that might be good --- for instance a drug that blocks both β and α receptors might be a good anti-hypertensive agent.
    - For instance, many beta blockers bind to both β_1 and β_2 receptors. These drugs can cause bronchiolar constriction at beta-2 receptors in the bronchi as well as slowing of heart rate at beta-1 receptors at the SA node in the heart.
  - Conversely, activity at a second receptor might be undesired—for instance a β blocker might also block serotonin receptors and cause bad dreams.



Mechanisms of Receptor Regulation

MECHANISM	DEFINITION
Tachyphylaxis	Repeated administration of the same dose of a drug results in a reduced effect of the drug over time
Desensitization	Decreased ability of a receptor to respond to stimulation by a drug or ligand
Homologous	Decreased response at a single type of receptor
Heterologous	Decreased response at two or more types of receptor
Inactivation	Loss of ability of a receptor to respond to stimulation by a drug or ligand
Refractory	After a receptor is stimulated, a period of time is required before the next drug–receptor interaction can produce an effect
Down-regulation	Repeated or persistent drug–receptor interaction results in removal of the receptor from sites where subsequent drug–receptor interactions could take place

Drug-Receptor interaction

- In most cases, a drug (D) binds to a receptor (R) in a **reversible bimolecular reaction**

$$D + R \underset{k_{\text{off}}}{\overset{k_{\text{on}}}{\rightleftarrows}} DR \underset{k_{\beta}}{\overset{k_{\alpha}}{\rightleftarrows}} DR^*$$

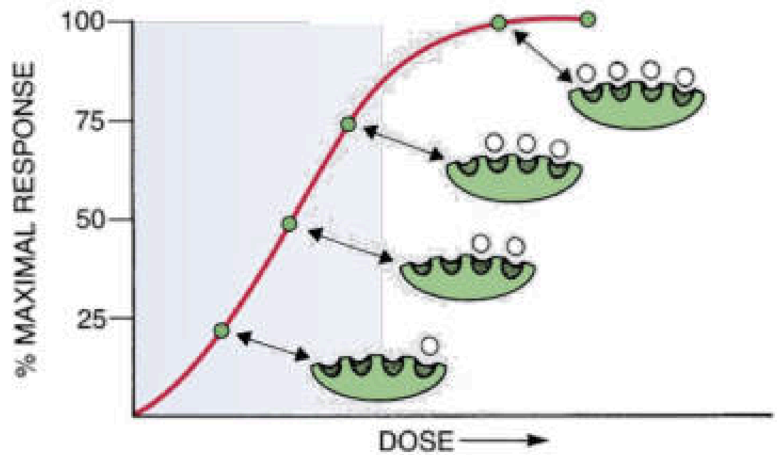
Potency **Efficacy**

- Antagonists can bind to the receptor and occupy its binding site and, therefore, participate only in the first equilibrium.
- Agonists, can bind, have the appropriate structural features to force the bound receptor into an active conformation (DR*).
- This conformational change leads to a series of events causing a cellular response.

Drug-receptor interaction

- Generally the intensity of response increases with dose
- Drug receptor interaction obeys the law of mass action

$$E = \frac{E_{max} \times [D]}{K_D + [D]}$$

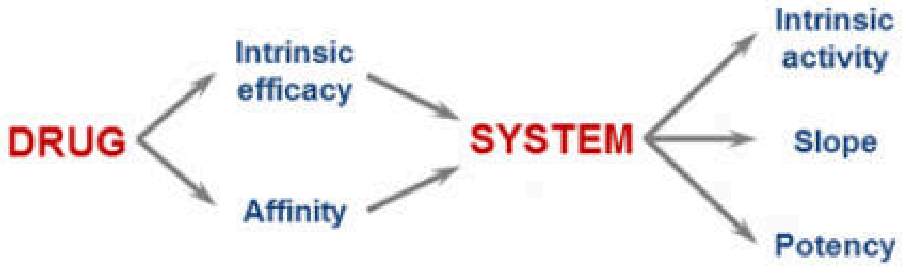


- E is observed effect at dose [D] of a drug
- E_{max} is the maximal response
- K_D is the dissociation constant of a drug receptor complex
- K_D is usually equal to the dose of a drug at which half maximal response (EC50) is produced (equilibrium phase)

Assessment of Receptor Occupation

Measure of Affinity

- **K_D** is a relative measure of **affinity** of a drug for its receptor.
- **K_D**; It varies **inversely** with the affinity of the drug for its receptor
- **High-affinity drugs** have **lower K_D values** and occupy a greater number of drug receptors than drugs with lower affinities.



Affinity & Intrinsic activity

- **Affinity:** It is the ability of a drug to bind with receptor (just bind)
  - The **affinity** of a drug for its receptors is reflected in **its potency**
  - Drugs with high affinity can bind to their receptors when present in low concentrations and have greater potency
- **Intrinsic activity:** It is the ability of a drug to activate a effector following receptor occupation.
  - The **intrinsic activity** of a drug is reflected in its **maximal efficacy**
  - Drugs with high intrinsic activity have high maximal efficacy because they are able to cause intense responses by causing intense receptor activation
  - Antagonists have no intrinsic activity.
  - Antagonists block the activity of the endogenous ligand by occupying the receptor and preventing the endogenous ligand binding

Graded Dose-Response Relationships:

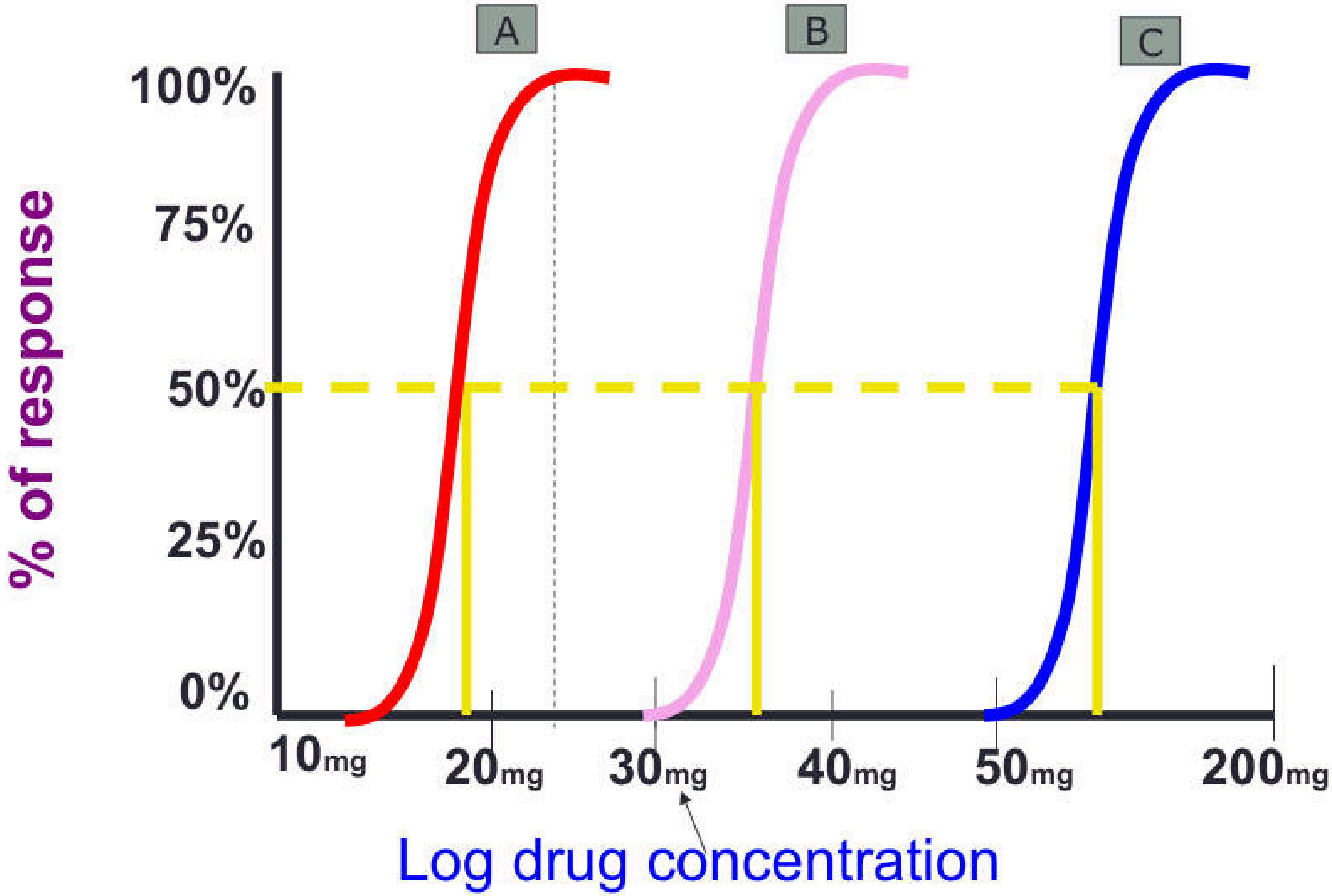
- When the response of a particular receptor-effector system is measured against increasing concentrations of a drug, the graph of the response versus the drug concentration or dose is called a

  “**graded dose-response curve**”
- Plotting the same data on semilogarithmic axes usually results in a sigmoid curve, which simplifies the MATH manipulation of the dose-response data .
- The Efficacy (Emax) and Potency (EC_{50}) parameters are derived from these data.
- The smaller the EC_{50}, the greater the potency of the drug.

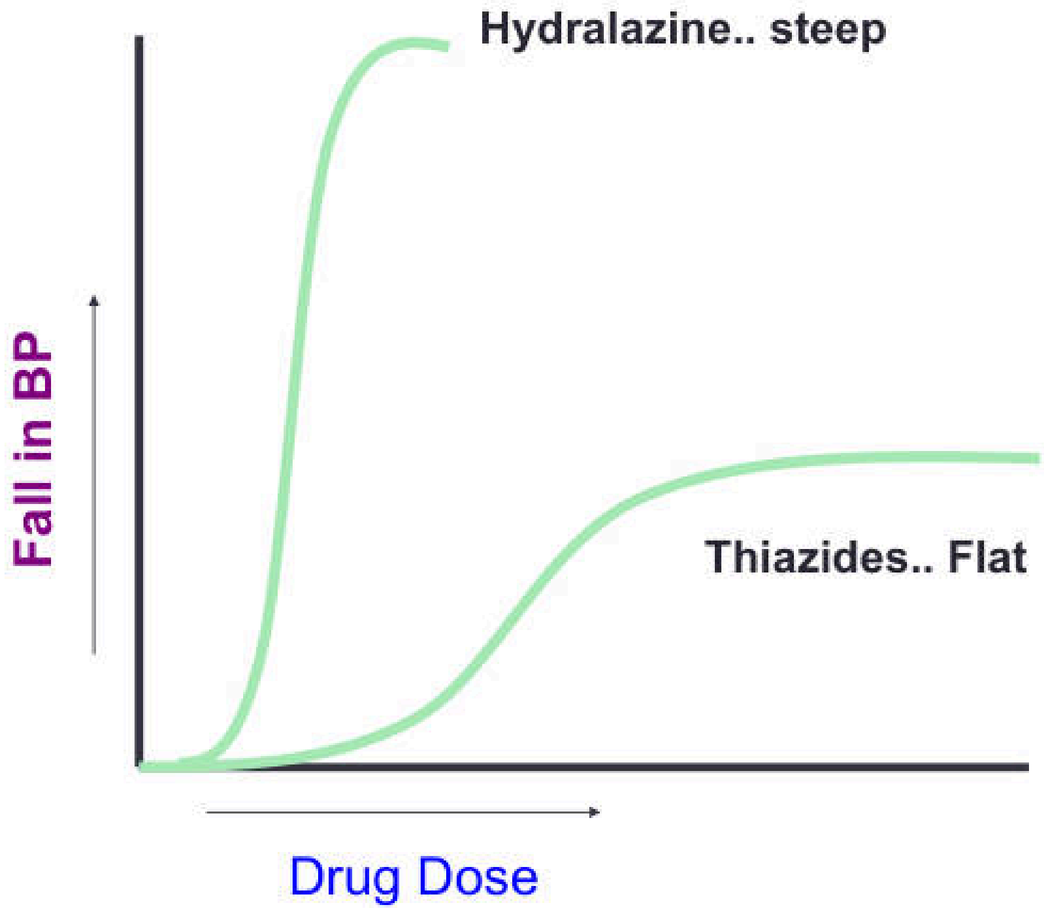
Efficacy and Potency

- Efficacy is the maximal response produced by a drug
- It depends on the number of drug-receptor complexes
- Potency is a measure of how much drug is required to elicit a given response
- The lower the dose required to elicit given response, the more potent the drug is
- EC_{50}: It is the dose of the drug at which it gives 50% of the maximal response
- A drug with low EC_{50} is more potent than a drug with larger EC_{50}

Potency of Drug A >Drug B > Drug C



Slope of DRC

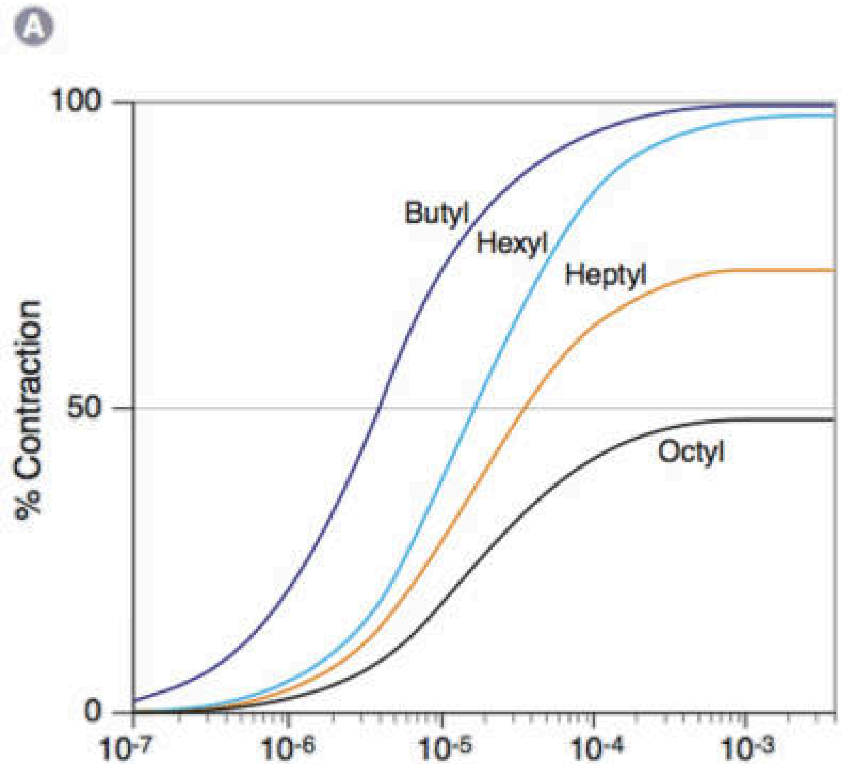


- The slope of midportion of the DRC varies from drug to drug
- A steep slope indicates small increase in dose produces a large change in response

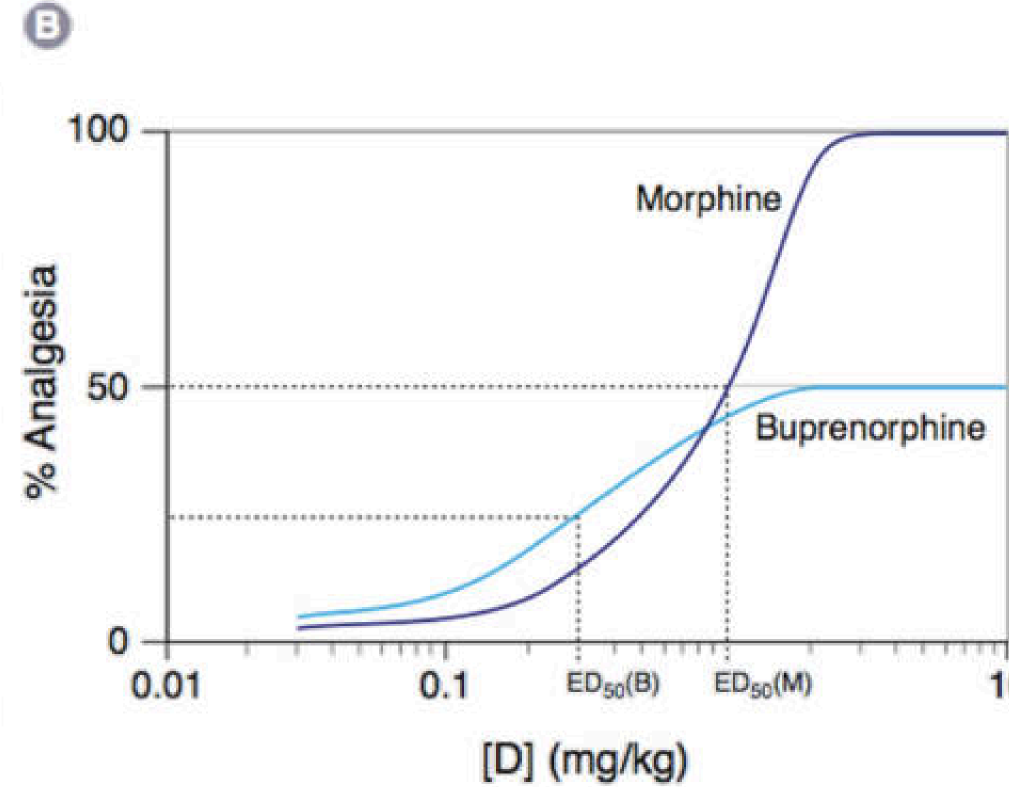
Intrinsic Activity
Agonists and Partial Agonists

- Intrinsic activity - The ability of a drug to activate a particular receptor when bound to it.
- Endogenous ligands have full (100%) intrinsic activity.
- Full agonists mimic the endogenous ligand and have full intrinsic activity.
  - Ex. meperidine in the previous example
- Partial agonists (partial antagonist or mixed agonist-antagonists) have less than full intrinsic activity (they are less efficacious than full agonists).
  - Ex. pentazocine

Intrinsic Activity: Agonists and Partial Agonists



Various alkyl derivatives of trimethylammonium all stimulate muscarinic acetylcholine (ACh) receptors to cause muscle contraction in the gut, but they produce different maximal responses, even when all receptors are occupied.



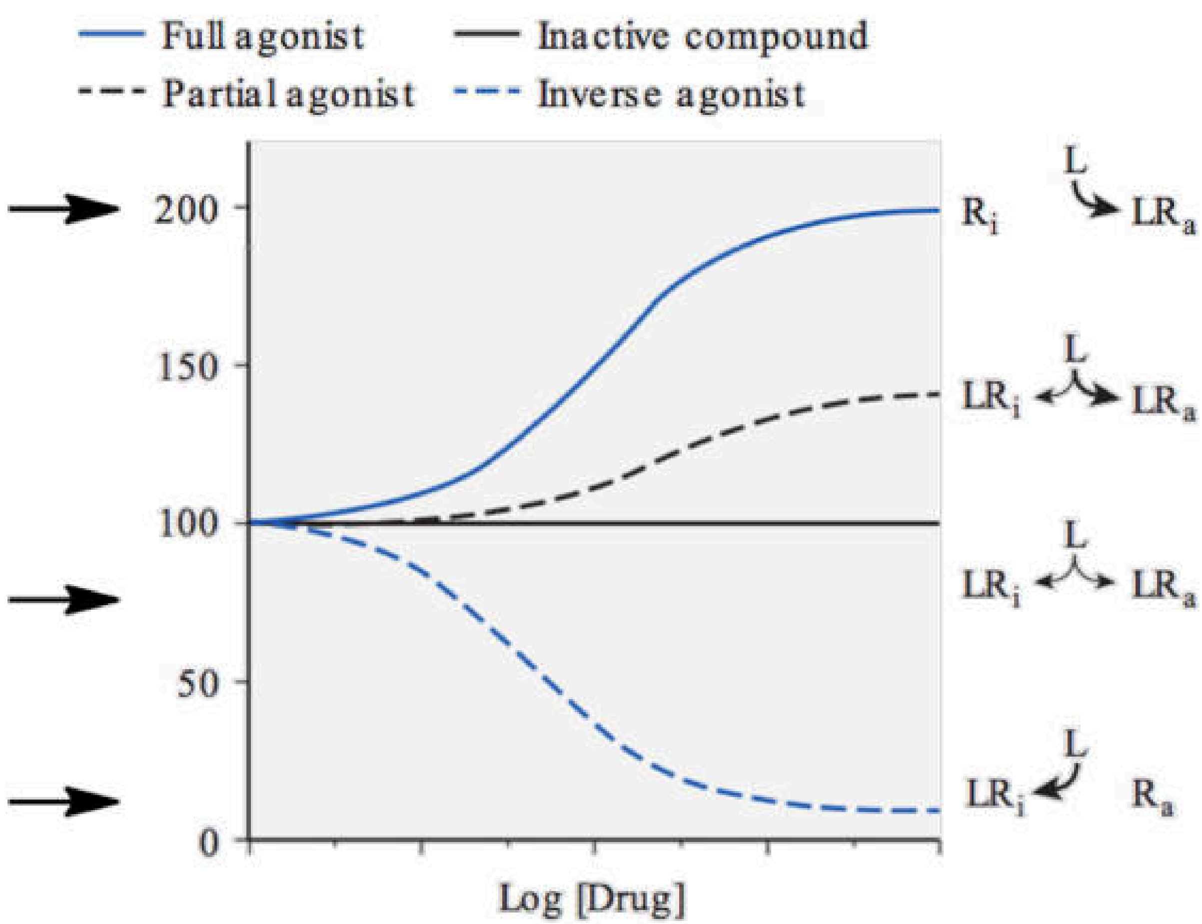
Partial agonists may be more or less potent than full agonists. In this case, buprenorphine (ED50 = 0.3 mg/kg) is more potent than morphine (ED50 = 1.0 mg/kg), although it cannot achieve the same maximal response as the full agonist.

Inverse Agonist vs. Agonist

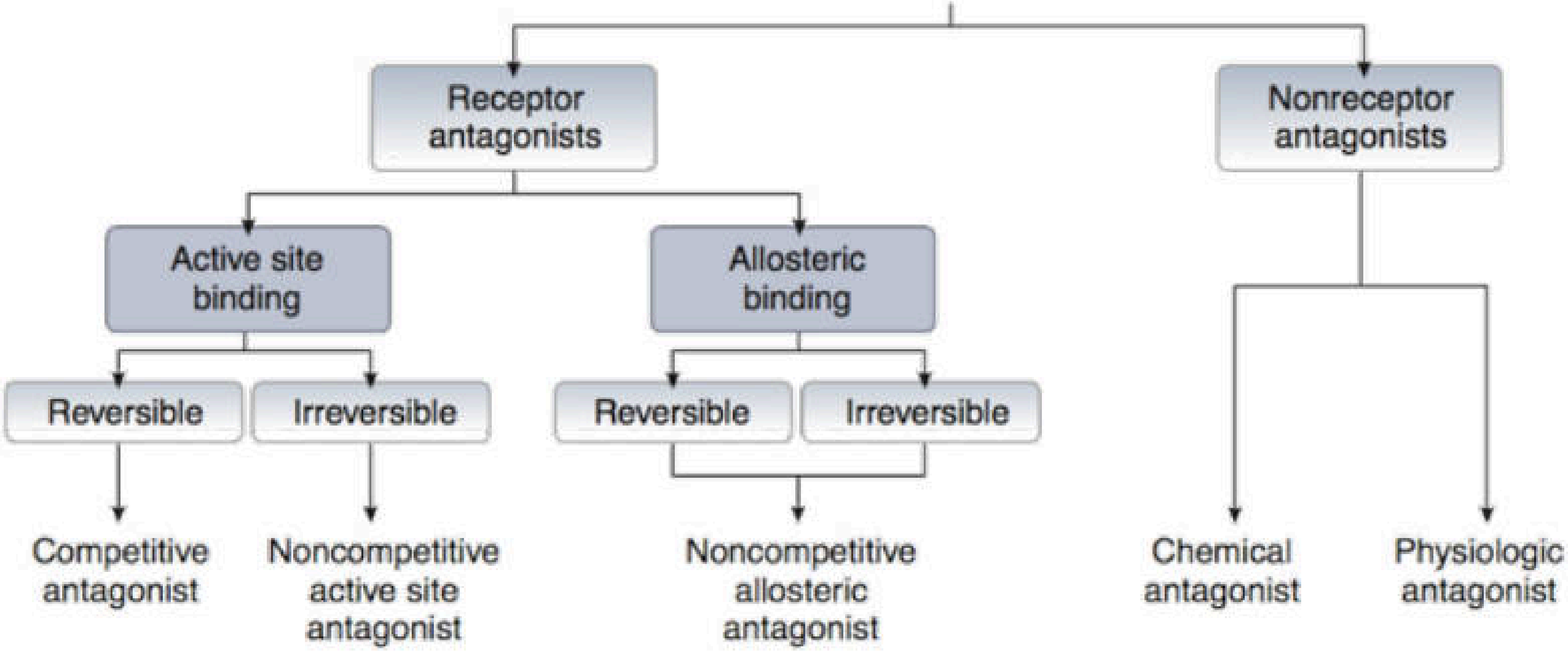
Inverse agonist: when a drug activates a receptor to produce an effect in the opposite direction to that of the agonist

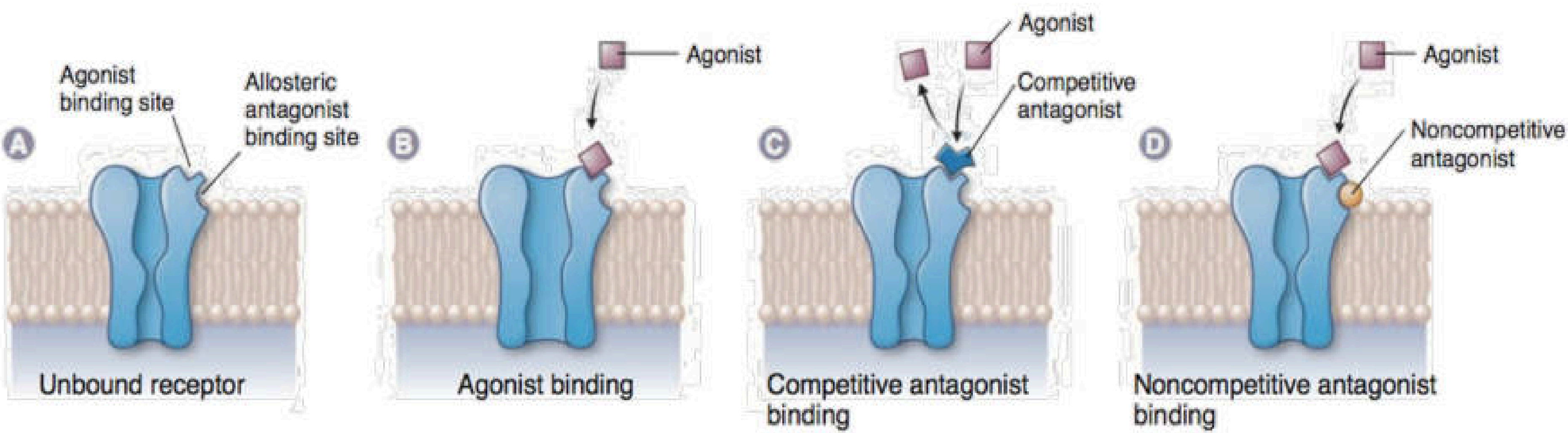
activation by saturating endogenous ligand

constitutive activty of receptor in absence of ligand

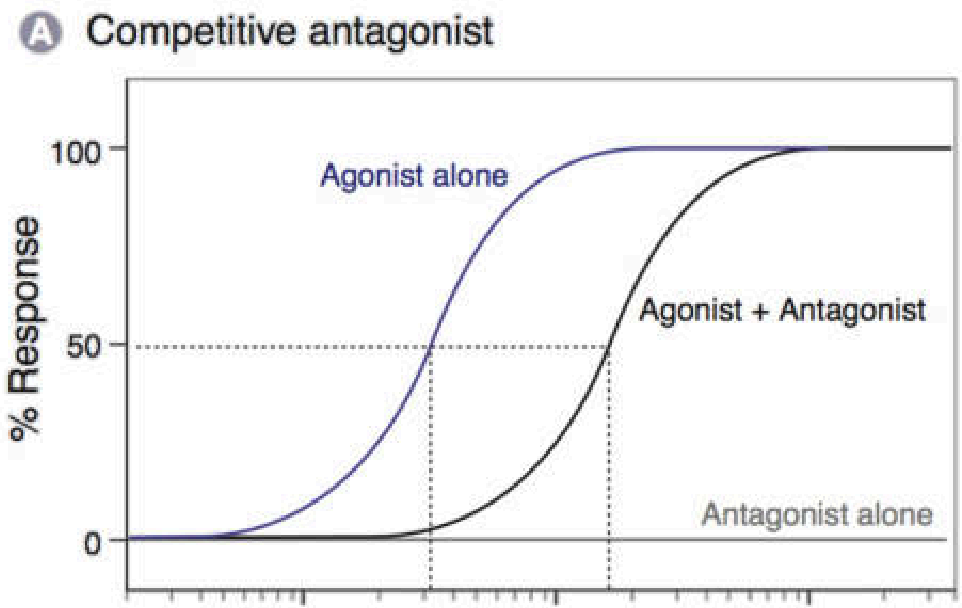


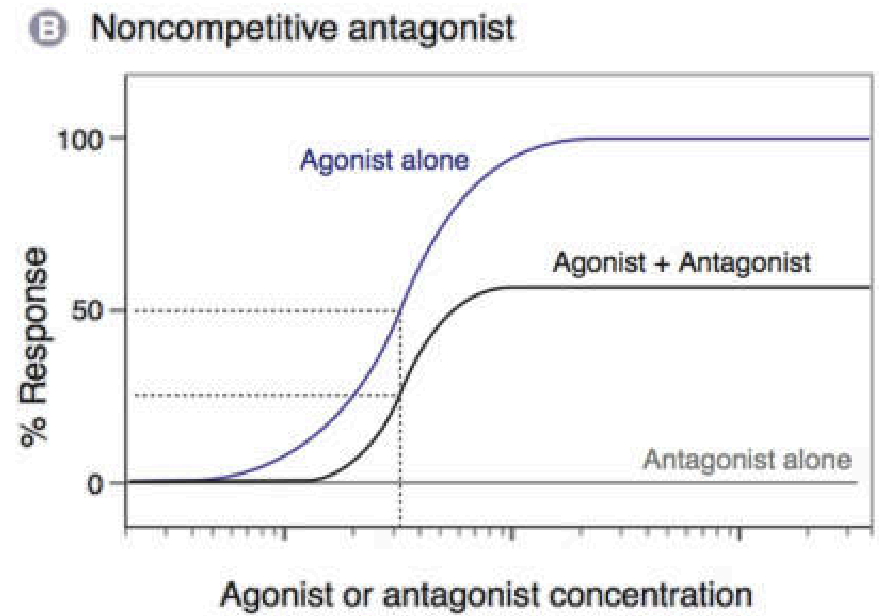
Type of Antagonists





Noncompetitive vs. Competitive Antagonism





<u>**Reduce Potency**</u>

<u>**Reduce Efficacy**</u>

Competitive (Surmountable) Antagonism

- More common type of antagonist
- Bind reversibly to receptors
  - Produce receptor blockage by competing with agonists for receptor binding
- If an agonist and an antagonist have equal affinity for a receptor, it will occupied by whichever is in a higher concentration
- Because binding is reversible, the inhibition that competitive antagonists cause can be surmountable by adding more agonists

Noncompetitive (Insurmountable): Antagonism

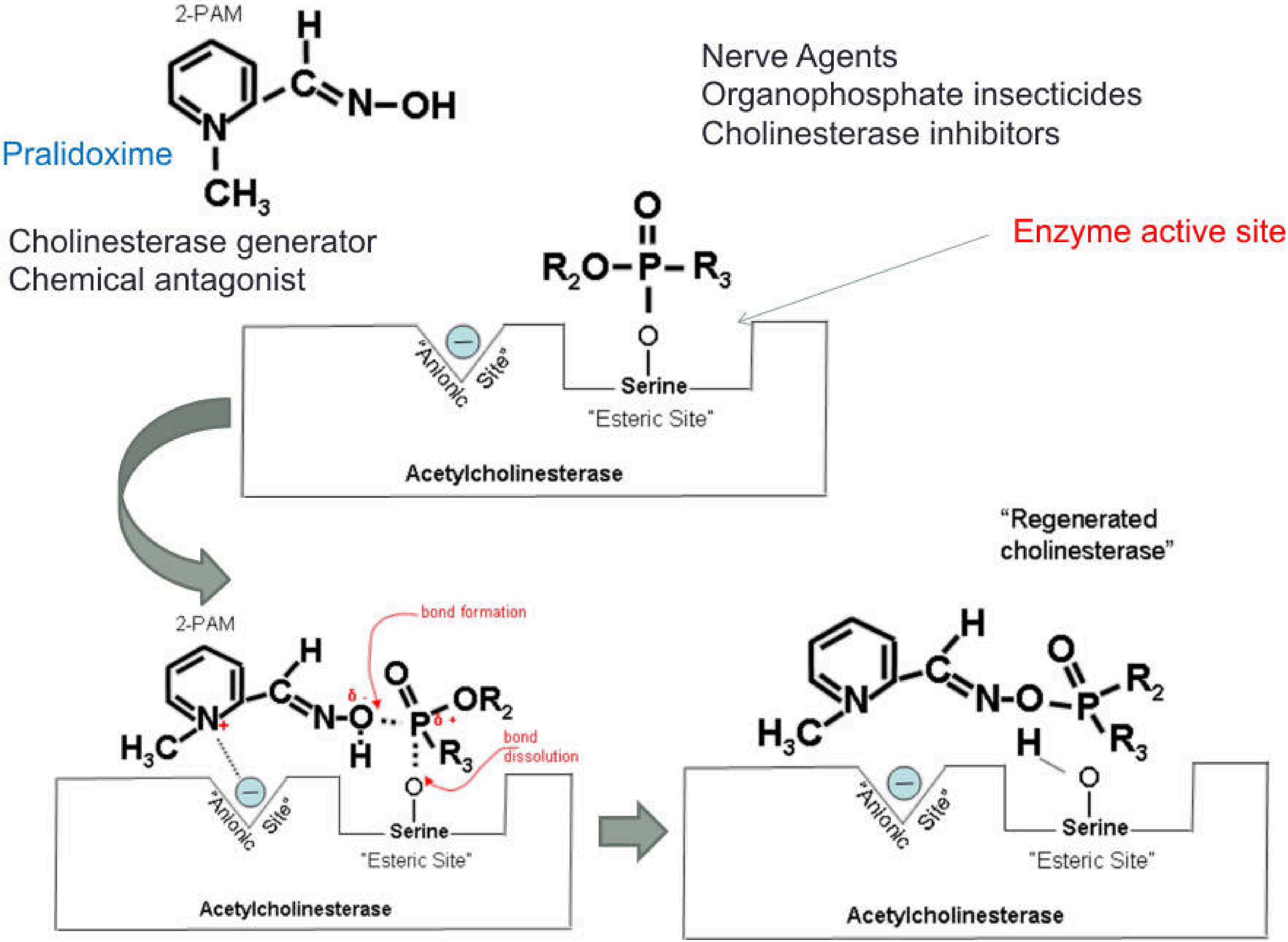
- Bind irreversibly to receptors
  - Reduces the total number of receptors available for activation by an agonist
- Because the binding of noncompetitive antagonists is irreversible, inhibition by these agents cannot be overcome, no matter how much agonist is given
- Noncompetitive antagonists are rarely used therapeutically because they are irreversible
- The effects of noncompetitive antagonists wear off as the receptors to which they are bound are replaced
- The effects may subside in a few days

Physiologic Antagonists:

- A physiologic antagonist is a drug that binds to a **different receptor**, producing an **effect opposite** to that produced by the drug it is antagonizing.
- Thus it differs from a pharmacologic antagonist, which interacts with the same receptor as the drug it is inhibiting.
- A common example is the antagonism of the bronchoconstrictor action of histamine (mediated at histamine receptors) by epinephrine's bronchodilator action (mediated at beta adrenoceptors).

Chemical Antagonists:

- A chemical antagonist is a drug that interacts directly with the drug being antagonized to remove it or to prevent it from reaching its target.
- A chemical antagonist does not depend on interaction with the agonist's receptor (although such interaction may occur).
- A common example of a chemical antagonist is dimercaprol, a chelator of lead and some other toxic metals.
- Pralidoxime, which combines avidly with the phosphorus in organophosphate cholinesterase inhibitors, is another type of chemical antagonist.


2-PAM
Pralidoxime
Cholinesterase generator
Chemical antagonist
Nerve Agents
Organophosphate insecticides
Cholinesterase inhibitors
Enzyme active site
"Anionic Site"
Serine
"Esteric Site"
Acetylcholinesterase
2-PAM
bond formation
bond dissolution
"Anionic Site"
Serine
"Esteric Site"
Acetylcholinesterase
"Regenerated cholinesterase"
"Anionic Site"
Serine
"Esteric Site"
Acetylcholinesterase

Summary of Agonist and Antagonist

Summary of Agonist and Antagonist Action

CLASSES OF AGONISTS

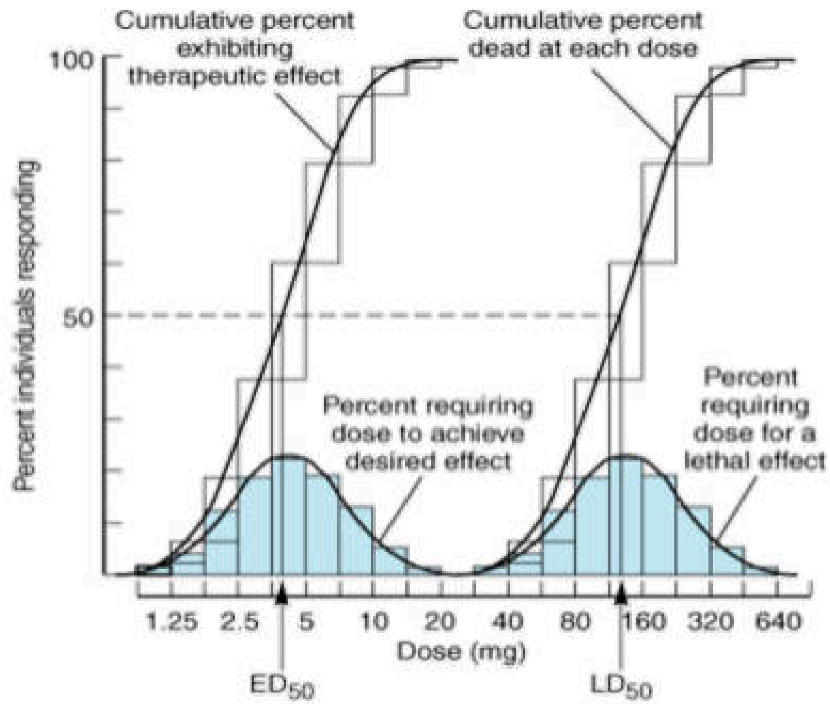
AGONIST CLASS	ACTION
Full agonist	Activates receptor with maximal efficacy
Partial agonist	Activates receptor but not with maximal efficacy
Inverse agonist	Inactivates constitutively active receptor

CLASSES OF ANTAGONISTS

ANTAGONIST CLASS	EFFECTS ON AGONIST POTENCY	EFFECTS ON AGONIST EFFICACY	ACTION
Competitive antagonist	Yes	No	Binds reversibly to active site of receptor; competes with agonist binding to this site
Noncompetitive active site antagonist	No	Yes	Binds irreversibly to active site of receptor; prevents agonist binding to this site
Noncompetitive allosteric antagonist	No	Yes	Binds reversibly or irreversibly to site other than active site of receptor; prevents conformational change required for receptor activation by agonist

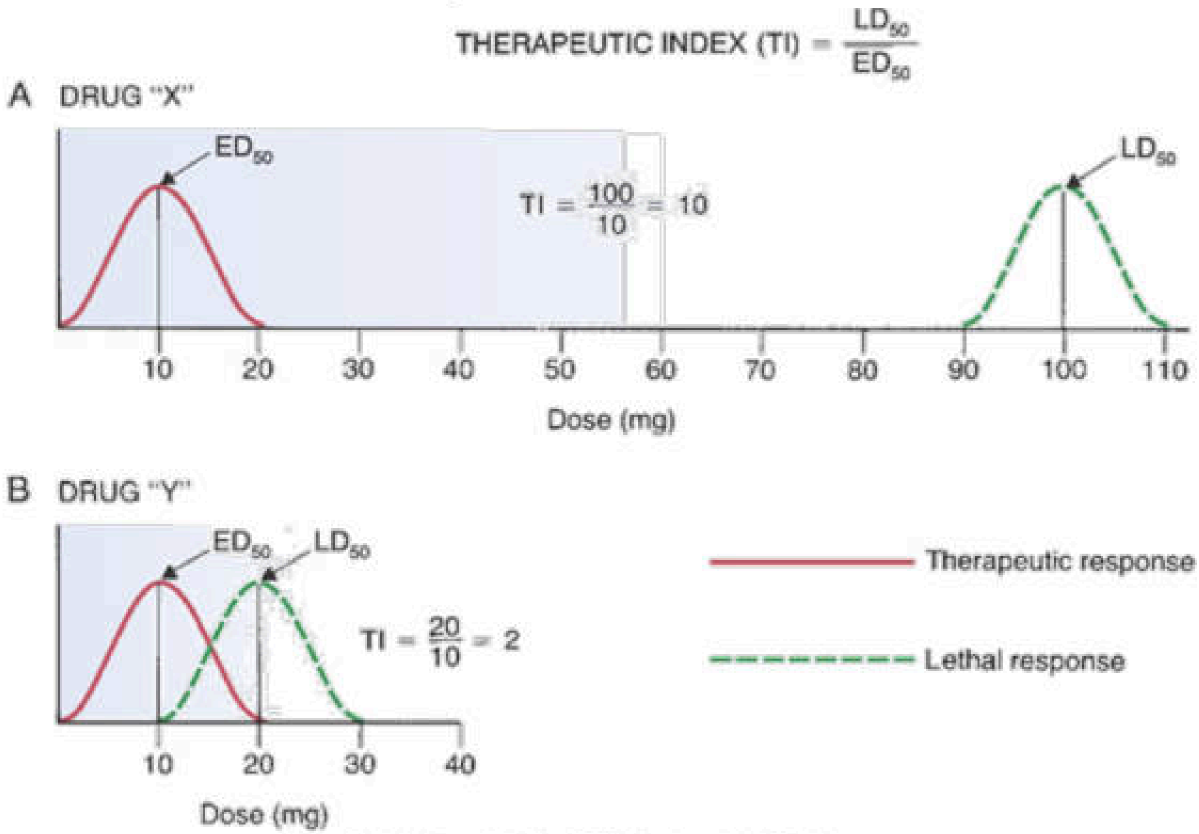
Quantal dose response curves

- The quantal dose-effect curve plots the **fraction of the population** that responds to a given dose of drug as a function of the drug dose.
- Similarly, the dose required to produce a particular toxic effect in 50% of animals is called the median **toxic dose (TD_{50})** If the toxic effect is death of the animal, a median **lethal dose (LD_{50})** may be experimentally defined



Quantal dose-effect curves are used to generate information regarding the margin of safety (Therapeutic index)

Therapeutic Index



- Drugs with a **low TI** should be used with caution and needs a periodic monitoring (less safe) e.g. warfarin, digoxin, theophylline
- Drugs with a **large TI** can be used relatively safely and does not need close monitoring (highly safe) e.g. penicillin, paracetamol
- Other terms used: wide safety margin, narrow safety margin

Others Drug Actions

Synergism

Two types:

1. Additive effect: the effect of two drugs are in the same direction and simply add up.
   - Effect of drug A + B = effect of drug A and B
2. Supraadditive effect (potentiation): the effect of combination is greater than the individual effect of the components.
   - Effect of drug A + B > effect of drug A + effect of drug B

Drug Interactions

- Drug-Drug Interactions
  - Intensification: Effect and/or Adverse Effects
  - Reduction
- Food-Drug Interaction
  - Absorption
  - Metabolism
  - Toxicity
  - Action
- Food-Herb Interactions

Dose, Dosing interval,
Route of Administration

Purpose of Action

Stimulation
Depression
Replacement
Cytotoxic action

Pharmacokinetics; PK

Absorption
Distribution
Metabolism
Elimination

Pharmacodynamics; PD

On-Off target

Biological Systems

Human
Microorganism
Tissue

Specific

DNA
Microbial organelles
Target macroproteins

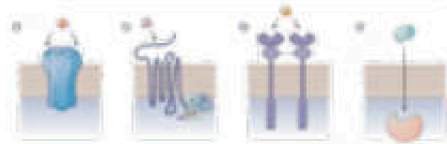
Non-specific

Graded dose-response

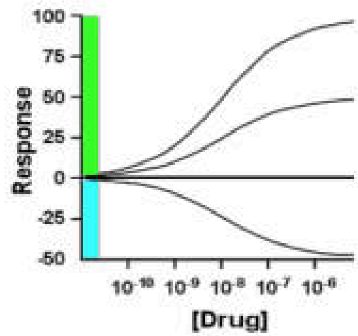
Intrinsic Activity

$$D + R \underset{k_{off}}{\overset{k_{on}}{\rightleftarrows}} DR \underset{k_{\beta}}{\overset{k_{\alpha}}{\rightleftarrows}} DR^{*}$$

Potency Efficacy



Agonist/Antagonist



Full agonist
Partial agonist
Neutral antagonist
Inverse agonist

Receptor antagonists

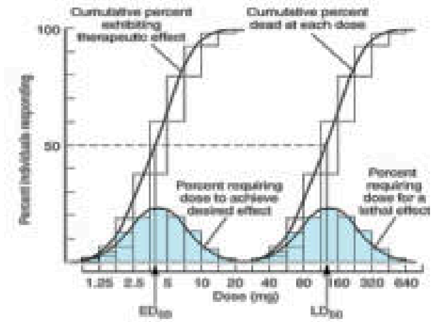
Reversible (Competitive/Surmountable)
Irreversible (Noncompetitive/Insurmountable)

Nonreceptor antagonists

Chemical
Physiological

Guantal dose-response

Therapeutic Index



Low or Large TI

Variability

Precision Medicine

Principle of Drug Actions